Dissertation

A Matter of Perspective: Unpacking the Effect of Different Variables on a Virtual Reality Game

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Context and Objective

Virtual reality has long promised to be an alternate dimension so convincing that the brain and body forget the mediation; it is, indeed, able to make players feel like they are inside the virtual environment, but it might also make them swear they will never touch a head-mounted display ever again. So, if the game is the same, what causes one or the other reaction?

The present work aims to design and develop a virtual reality game with personalizable parameters (e.g., setting [indoor vs. outdoor], soundscape [quiet vs. noisy], scene [calm vs. chaotic], character behaviour [passive vs. aggressive] and the visual configuration [field of view, resolution, brightness, motion blur, etc.]) that can be manipulated across sessions to understand how such variables affect the player experience.



I Expect You To Die



Out of Space



Overcooked

Examples of games that illustrate how the same mechanic, replayed under different conditions (e.g., environment or pacing), produces different gameplay experiences.

Work Plan

- Review the state of the art;
- Define the personalizable parameters and design the game concept;
- Iteratively develop, playtest and refine the game;
- Evaluate the game under different configurations;
- Write the dissertation document.

Recommended Elective Course:

- Realidade Virtual e Aumentada

Observations

The work will take place at IEETA, in the Virtual and Augmented Reality Laboratory.